

AI-Powered Candidate Ra

Beyond Keywords: Semantic Hiring Intelligence

A High-Performance Hiring Pipeline for Modern Recruiting Teams

Technology stack: Sentence-Transformers | Google Gemini | RapidFuzz | FPDF2

The Problem

Why traditional screening methods consistently fail recruiting teams

- Keyword Matching misses rockstars: Legacy applicant tracking systems (ATS) look for exact matches. A candidate with 'Deep Learning' experience might be filtered out for a job listing requiring 'Neural Networks'.
- Recruiter Burnout & Subjectivity: Manually reviewing hundreds of complex engineering resumes is tedious. Recruiters may miss qualified candidates due to unconscious bias, and missed high-potential hires.
- Isolated Visual Footprints: Resumes contain dense text but completely ignore real-world impact. A candidate's GitHub repository or LinkedIn technical authority is ignored.
- No Trajectory context: Traditional screening lacks the ability to analyze growth trajectory and potential for future roles.

Our Solution: Multi-Signal Hybrid Scoring

A cohesive, dual-engine screening pipeline

1. Input Datasets

- Job Description (.txt)
- Candidate Profiles (.csv)
 - * Full resume summaries
 - * Years of experience
 - * Core skills
 - * GitHub & LinkedIn activity



2. Hybrid Scorer (100%)

- Semantic Score (40%)
 - * Sentence embeddings, Cosine Sim
- Skills Score (25%)
 - * Set Intersection + RapidFuzz
- Experience Score (20%)
 - * Years vs Target + Title Fuzzy Sim
- Behavioral Score (15%)
 - * Open source & Social signals

Signal 1: Semantic Similarity

Unlocking contextual understanding with sentence embeddings

- Dense Vector Mapping: Leverages the 'all-MiniLM-L6-v2' Sentence-Transformer model to map text into high-dimensional vector spaces.
- Beyond Keywords: Captures the semantic intent of candidate summaries, projects, and JD requests. Example: JD requests 'Large Language Models' and a candidate writes 'expert in fine-tuning LLMs' - high-similarity contexts.
- Cosine Similarity Scoring: Computes the directional cosine similarity between the JD and candidate vectors, outputting a 0 - 100 score.
- Eliminates Formatting Bias: Normalizes resume text before embedding, focusing purely on semantic content.

Signal 2: Fuzzy Skills Matching

Robust technical skill evaluation with RapidFuzz

- Set Intersection + String Metrics: Combines hard exact set matching with fuzzy character matching to identify relevant skills across candidate technologies.
- Resolves Synonyms & Typos: Leverages 'rapidfuzz' (token sort ratio) to cleanly match variations, bridging the gap between 'TF' and 'TensorFlow', 'scikit-learn' and 'sklearn', or 'vector d' and 'vector search'.
- Target Skills Library: Validates candidate skill sets against 30+ highly desired domain-specific skills, including Frameworks (PyTorch, JAX), Infrastructure (Kubernetes, Triton), and Databases (Pinecone, Redis).
- Balanced Normalization: Rather than requiring a candidate to possess every skill, the system weights key core skills, allowing highly specialized profiles to achieve perfect 100/100 marks.

Signal 3: LLM Recruiter Assessment

Google Gemini structured logic for top-tier candidates

- Pre-filtered Focus: The pipeline pre-screens all candidates and calls Google Gemini (1) to assess fit, (2) optimize prompt usage, manages API costs, and speeds up execution to under 15 seconds.
- Multi-Dimensional Recruiter Prompts: Gemini evaluates the candidate's complete resume against (1) Technical Skills & Experience, (2) Professional Growth Trajectory, (3) Cultural & Behavioral Signals.
- Guaranteed Structured Response: Utilizes structured outputs (response_mime_type="application/json") to return standardized numeric scores and a qualitative reasoning sentence.
- Explainable AI: The generated LLM reasoning is saved directly in the final CSV, giving hiring managers context on why a candidate was ranked high or bypassed.

Signal 4: Behavioral & Activity Signals

Looking beyond the CV to real-world impact

- GitHub Open Source Impact: Scans candidate profiles for indicators of code quality and community involvement, identifying keywords like 'PyTorch contributor', 'developer of LangChain', and simulates stars metrics.
- LinkedIn Professional Engagement: Assesses social footprint and technical authority through metrics like professional follower count (e.g. 10k+ followers).
- Community Validation: Rewards candidates who actively share knowledge, mentor others, or contribute to open-source projects for engineering teams.
- Pragmatic Score Integration: Scores are blended together to create a 0-100 Behavioral Score, combining traditional technical skills with real-world activity.

Scoring Weights & Blended Formulas

Rigorous math ensuring robust, balanced rankings

Evaluation Signal	Weight	Focus Area & Methodology
Semantic Similarity	40%	Dense vector matching, context-aware resume alignment
Fuzzy Skills Match	25%	Exact matching + RapidFuzz (Jaro-Winkler/Token ratio)
Experience Relevance	20%	Quantitative years of experience vs Target (5.0 yrs) + 1
Behavioral Signals	15%	Simulated digital footprint (GitHub contributions & Linke

Final Blended Scoring Equation (for Top 20 Candidates):

Final Score = 60% * [Hybrid Score] + 40%

**Note: If Gemini API key is missing, system gracefully degrades to Final*

Hackathon Pipeline - Top Candidate Shortlist

Real results from the Senior ML Engineer ranking run

Rank	Candidate Name	Current Professional Title	Exp (Years)
1	Amina Yusuf	Principal AI Architect	9.5
2	Rajesh Sharma	Lead Machine Learning Engineer	8.2
3	Dr. Emily Chen	Senior Research Scientist (AI)	7.8
4	Arjun Nair	Senior ML Infrastructure Engin	6.9
5	Devon Harris	Senior AI Platforms Engineer	9.1

**Pipeline successfully processed 50 synthetic candidates. Output saved to 'output/ranked_candidates.csv'.*

Why This Works

The strategic business advantages of semantic recruiting

- 10x Screening Speed: Instantly ingests, normalizes, embeds, and ranks hundreds of resumes, slashing Time-to-Interview.
- Minimizes Manual Bias: Ensures every candidate is measured objectively using semantic context, bypassing names or formatting visual biases.
- Optimal API Cost Design: By utilizing sentence-transformers locally and calling Gemini for semantic intelligence while minimizing API token bills by 80%.
- Recruiter-First Focus: Provides immediate, clear explanations ('llm_reasoning') for suggestions, enabling highly-informed interviewing decisions.

Empowering Recruiting Teams with Semantic Intelligence